

QT 407: Introduction to Quantum Sensing

Course Code: QT 407

Course Name: Introduction to Quantum Sensing

L-T-P-C: 3-0-0-3

Prerequisites: Engineering Mathematics (Linear Algebra, Complex algebra, basics of 2nd of ODEs and initial value problems, 2nd order PDEs and boundary value problems, Probability and Statistics, Random variables). Maxwell's equations and EM theory at the level of the core physics syllabus from AICTE model

Intended for: UG/PG/PhD

Distribution: Elective PG/Elective UG

Semester: Odd/Even

Approval: 47th Senate

Preamble

Students of this course learn (i) The basics of classical sensing, (ii) Aspects of quantum measurement, (iii) Ways to quantify quantum sensing, (iv) About measurements of quantum states of light, (v) About the applications of quantum sensing

Course Content and syllabus

- **Classical sensing**
 - Photo detection
- **Classical noise**
 - Johnson Noise, Telegraph noise, flicker or $1/f$ noise
- **Sensitivity of classical measurements**
 - Classical Fisher information
 - Cramer - Rao bounds (information theory basics may be required here).
- **Quantum measurements**
 - projective/orthogonal measurements
 - Approximate/non-orthogonal measurements
 - Weak continuous measurements
 - Error-disturbance relations
 - Standard quantum limits
 - Quantum non-demolition measurements
- **States of light**
 - Fock states
 - Coherent states
 - Squeezed states

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- Tomography
 - Wigner quasi-probability distribution
 - P-distribution
 - Husimi Q function
 - **Quantum photo detection**
 - Square-law detectors, Intensity measurements and Photo-detection
 - Linear Detectors and Quadrature Measurements
 - **Quantum Cramer-Rao bounds**
 - **Single photon-based sensing applications**
 - **Entanglement based sensing applications**
 - **Atomic state-based sensing, solid-state spin-based sensing applications (gravimetry, magnetometry)**

Course References

1. *Quantum Measurement and Control*, Howard Wiseman and David Milburn, Cambridge University Press (2014)
2. *Quantum Measurement*, Vladimir Braginsky and Farid Ya Khalili, Cambridge University Press (1995)
3. *Quantum Information Science* – Manenti R., Motta M., 1st Edition, Oxford University Press (2023)